

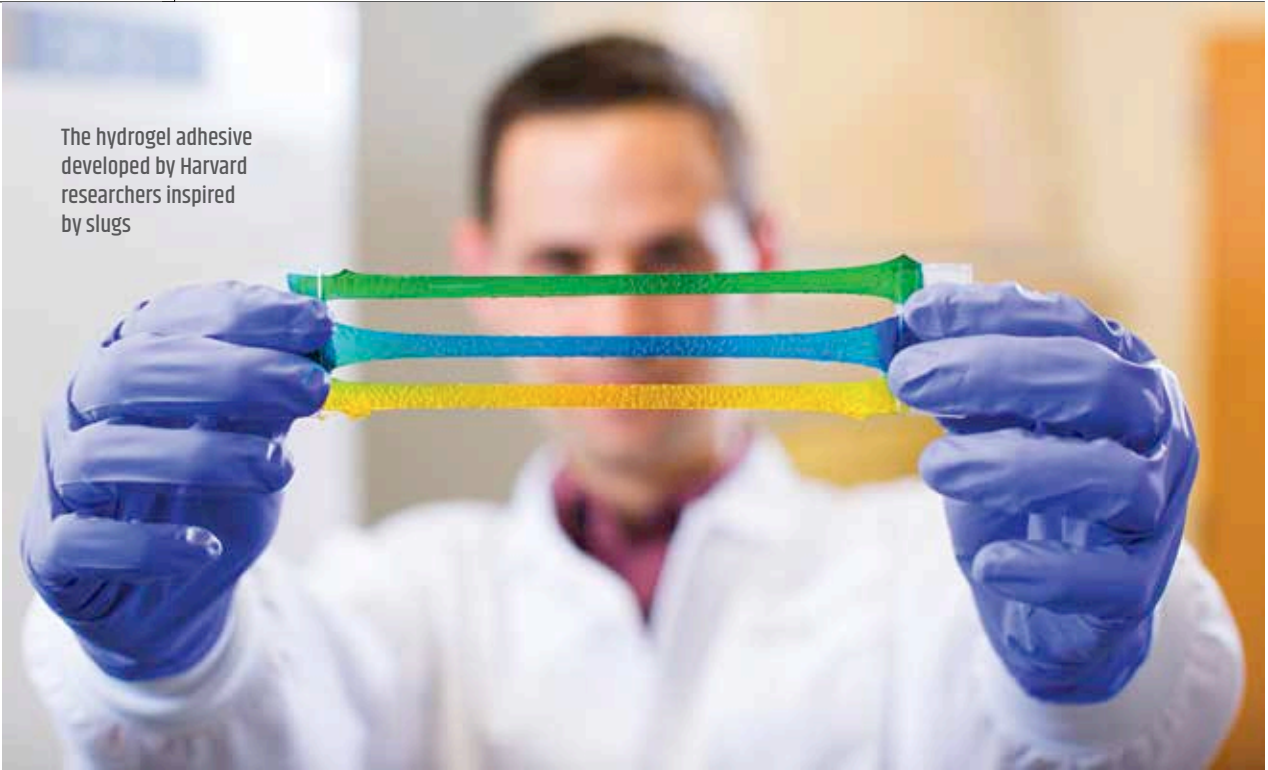
**MeTro hydrogel**

A video showing how the MeTro surgical hydrogel can be used on patients during a surgical operation. <https://digit.in/aug20-09>

**Tough gel adhesives**

A demonstration of the unique physical properties of the glue formulas developed at Harvard WySS. <https://digit.in/aug20-10>

The hydrogel adhesive developed by Harvard researchers inspired by slugs



Credit: Harvard WySS

MEDICAL ADHESIVES

A number of new adhesive substances are replacing the problematic staples and stitches used for holding damaged tissue together.

Aditya Madanapalle | aditya@digit.in

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ore than 300 million major surgeries are conducted around the world, every year. During these

surgeries, the surgeons have to use sutures to close the wounds or incisions which can themselves be problematic. They delay the healing of the damaged tissue and can cause infections. They can also cause pain, induce stress on the tissue and leave scars on the skin. To ease the healing process, researchers are coming up with new ways in which the wounds can be closed, which are superior

alternatives to staples, sutures, stitches or wires. In the hard to reach areas of the body, this is also a tricky process that requires an expert hand. For over thousands of years, there have been no innovations or improvements in this department. A suitable surgical glue would have to be fast acting, strong, easily removable, non toxic (or biocompatible), flexible and work with moving organs. It is only recently that such adhesives are being developed.

STICKY TAPE

Researchers from MIT have been developing a sticky tape made specifically for surgeries. The tape works through the wet surfaces of the internal tissues and can even be used on

sensitive and delicate organs like the heart. Unlike previous tissue glues, the tape works rapidly even on the wet surfaces. This glues offer tougher adhesion than the alternatives found in nature such as the ones used by barnacles or mussels to stick to underwater rocks. The tape is inspired by the sticky material used by spiders to make their webs. The spider's web works by removing water from a small patch of the skin of the insect stuck in the web, allowing the adhesion to form quickly. To absorb the water from the surface of the tissue, the researchers used polyacrylic acid, which is the same substance that is also used in diapers. As soon as the polyacrylic acid absorbs the water, another type of chemicals called NHS esters are used to stick the two surfaces together, through the formation of strong covalent bonds.

Then, the researchers had to make this glue last for long periods within